Bench 6 — ESP-IDF Filter Architecture Benchmark

Phase 6 comparative study over the clean ESP-IDF captures · 37 candidates · frozen configurations · corrected reconstruction

Hardware: 2× MS5611 GY-63 clones (foam-covered) + MPU6050, ESP32-S3 @ 240 MHz · firmware esp-idf/vario @ 8e79e68 · baro 40 Hz + 10 Hz temp, IMU 100 Hz

Status: complete frozen-configuration study with controls. **No estimator is selected or frozen.** This PDF is the DEF-014-corrected re-issue: the handtest reference-smoother boundary artifact is fixed and rankings are re-computed (§3.3, §6). Scripts: filter_benchmark/bench6/ (run6 → equiv6 → rate6 → tune6 → mcucpu6 → figures6); re-run on BMP581/LSM6DSV16X captures by registering the new stems in datasets.py.

1. Objective and datasets

Re-evaluate the full filter trade space on the clean ESP-IDF recordings — 29 standalone filters, stacked S1–S5, CV/Kalman references and IMU-fusion candidates — with the reconstruction defect DEF-001 corrected (fixed 101325 Pa ISA reference, firmware-exact compensation). This is a fresh comparative study, not a confirmation of previous rankings. Both barometers are evaluated independently throughout (the two clones differ; sid1 is consistently ≈1.15–1.4× noisier). Corrupted captures are excluded from scoring and retained as documented fault-test data.

Table 1. *Scoring datasets and quality verdicts (measured, not labelled).*

Capture	Role	Split	Quality
221124_idf_elevator_2x	elevator run 1: 4 legs ±38.2 m, –1.72/+1.66 m/s	train	CLEAN
221420_idf_elevator_2x	elevator run 2 (same profile)	validation	CLEAN (3× 20 ms gaps)
215709_idf_handtest3	static + gentle slow moves; 'toss' was DEF-014 artifact	train	GOOD (7× 20 ms gaps)
234521_idf_p0fix_static	static 75.3 s	validation	CLEAN
234944_idf_bbuf_check	static 20.2 s	validation	CLEAN
221109_idf_elevator_2x	aborted capture	excluded	USB-stall dropouts (DEF-008)
215027_idf_foam_static	foam static	excluded	CORRUPTED: 3703 seq gaps / 6192 dup IMU ts

Controls: frozen configurations first; tuning is a separate leakage-controlled pass (train-select, validation-report); a sampling-rate control precedes all cross-chain comparisons; MCU-vs-replay equivalence precedes all replay-vs-firmware comparisons. Estimator output and virtual-audio behaviour are scored separately.

2. MCU equivalence and new firmware-chain findings

Formal equivalence (queued task): PASS for D01. On the post-p0fix statics, with the era-correct chain (fixed 101325 Pa reference against the firmware's 10× pressure scale — DEF-011 — plus the boot d2 seed), the Python replay reproduces MCU tone velocities with steady-state bias ≤ 0.3 mm/s, p95 |error| = 1.0 mm/s and 100 % audio ON/OFF transition agreement (24 and 37 transitions, both sids). C03 agrees to p95 71–105 mm/s (its gain-cache quantization). S4/S5/FUS emit no steady audio transitions on statics — a vacuous, honestly reported result.

DEF-011 (new, major): baro_proc.c::compensate_pa returns p01mbar·10, ten times the common.py scale. Under the moving-p0 firmware the ratio cancelled; the p0fix hardcoded 101325 Pa (common-line units) against the still-10× pressure, so post-fix firmware velocities are scaled by exactly 10^0.1903 = 1.5498 and absolute altitude is garbage (–23 725 m). Self-consistent within the firmware, so bench captures remain valid; equivalence replays must use the isa_fw10 chain.

DEF-010 mechanism: the boot d2 seed (prom[5]·256, 20 °C) steps the compensation at the first real T frame → startup audio flap (tones clamp at +6.67 m/s), amplified to a –1.5×10■ m/s transient by the DEF-011 scale; emulated by seed_d2=True, and startup (t < 8 s) is always reported separately.

Gate-decision chaos (pre-p0fix moving window only): during the first 50 s of the elevator/handtest captures, S4/S5 (and mildly C03/FUS) replay tones diverge from MCU by seconds, while the plain

Kalman D01 — the core of both stacks — matches to ~1 mm/s everywhere. The raw reconstruction is therefore right; the divergence lives in discontinuous gate latches amplifying sub-LSB micro-differences while the defective moving reference was active. It affects only old-firmware equivalence bookkeeping, never scoring (fixed-ISA), and vanishes with the frozen reference (frozen-era p95: D01 1.0–1.2 mm/s, S4 1.1–2.1 mm/s, S5 1.0–1.3 mm/s, all captures).

Asymmetric tone clamp (new observation): implied v = (f–500)/450 with f ∈ [400, 3500] Hz represents v ∈ [–0.22, +6.67] m/s; the firmware suppresses tones below 400 Hz, so MCU tone streams are blind to strong sink. Replay audio metrics are the authoritative comparison for sink behaviour.

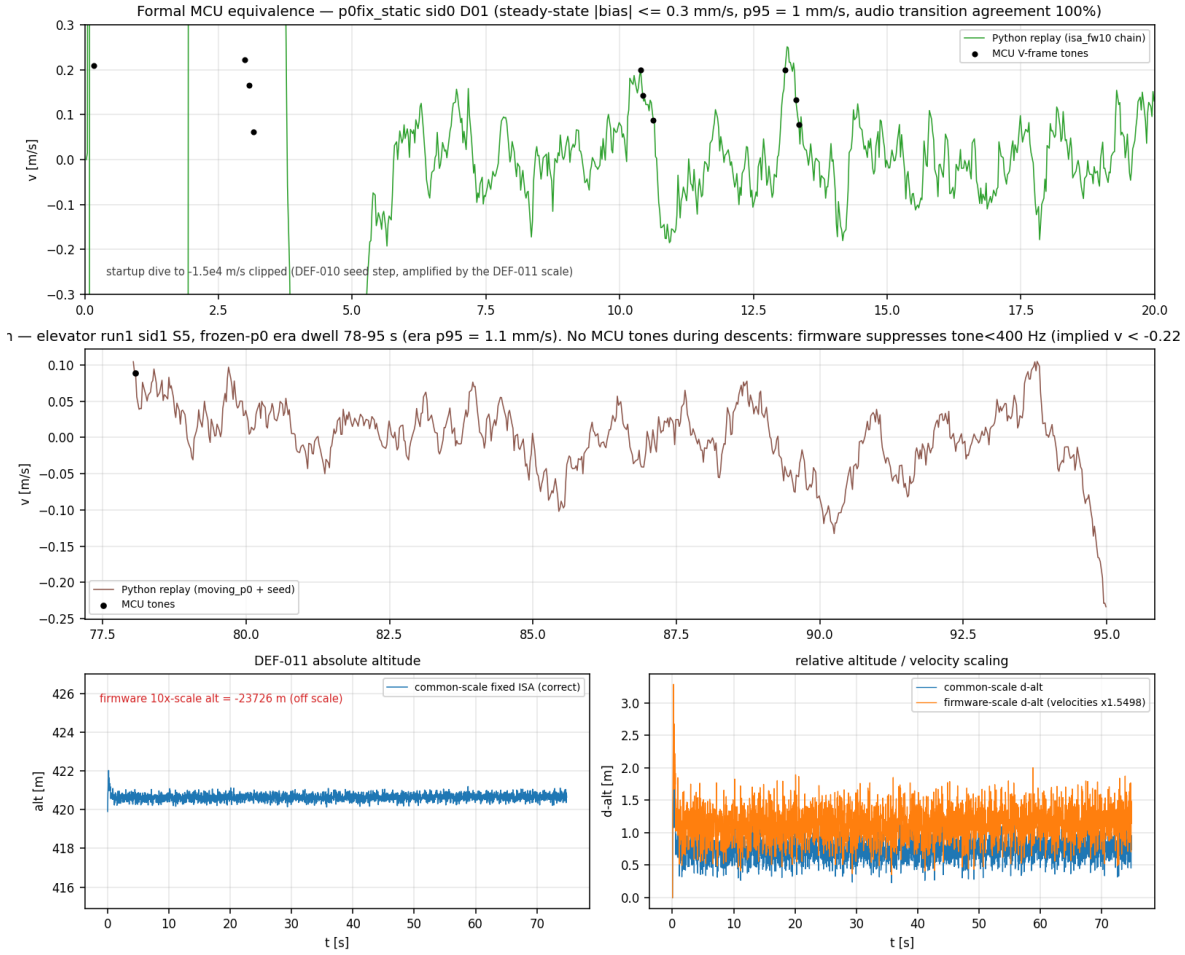


Figure 1. *Formal MCU equivalence (a), harness validation in the frozen-p0 era (b), and the DEF-011 pressure-scale defect (c).*

3. Frozen-configuration results

3.1 Static silence

Table 2. *Rest velocity RMS (m/s, medians). Phase 3/4 numbers were measured on the defective moving/session-mean reference — most of the apparent movement is the DEF-001 correction, not the filters.*

Candidate	Phase 3/4 rest RMS	Bench 6 rest RMS	Δ
B02_project_kalman	0.096	**0.0165**	–83 %
C02_AlphaBetaGamma	0.021	0.0217	—
C01_AlphaBeta	0.026	0.0282	—
S1_RobustAdaptiveKF	0.044	0.0346	–21 %
S5_LightGateKF	0.045 [M]	0.0369	–18 %
S4_FastSlowGated	—	0.0371	—
S2_DualTimescale	0.084	0.0373	–56 %
C03_CV_tracker	0.048	0.0485	—
D01_Kalman_CV	0.101	0.0567	–44 %
B02_Butter2	2.55	1.49 (rate artefact, §4)	—

Audio false-latch on statics: 0 latches/h for the whole bank; the known chatterers remain A03_Holt (~5615/h), S3 (~4450/h), FUS_Q10 (~2273/h), E06 (111/h), D04 (55/h); unadaptive baselines B00/E01–E03 chatter at 6400–6700/h.

3.2 Elevator controlled motion (run 1, medians over sids)

Table 3. *Sustained climb and descent track equally (\|bias\| ≤ 6 mm/s on ±1.7 m/s legs). The exit proxy is floored at ≈6.8 s by dwell wobble [D]. No filter missed any lift — elevator discrimination is saturated.*

Candidate	leg RMS	leg bias	rise [D]	exit [D]	dwell false s	missed
C03_CV_tracker	0.089	–0.006	1.27	6.53	2.18	0
S2_DualTimescale	0.091	–0.005	1.34	4.44	2.01	0
S4_FastSlowGated	0.091	–0.006	1.34	6.07	2.15	0
E06_RobustAdapt	0.092	–0.005	1.18	6.80	2.33	0
D01_Kalman_CV	0.093	–0.005	1.26	6.80	2.27	0
S1_RobustAdaptiveKF	0.097	–0.005	1.30	5.18	1.97	0
S5_LightGateKF	0.098	–0.005	1.31	6.28	2.13	0
C01_AlphaBeta	0.106	–0.004	1.67	3.30	2.21	0
FUS	0.241	–0.028	1.77	6.81	4.95	0
B02_Butter2	1.489	–0.006	n/m	6.91	19.9	0

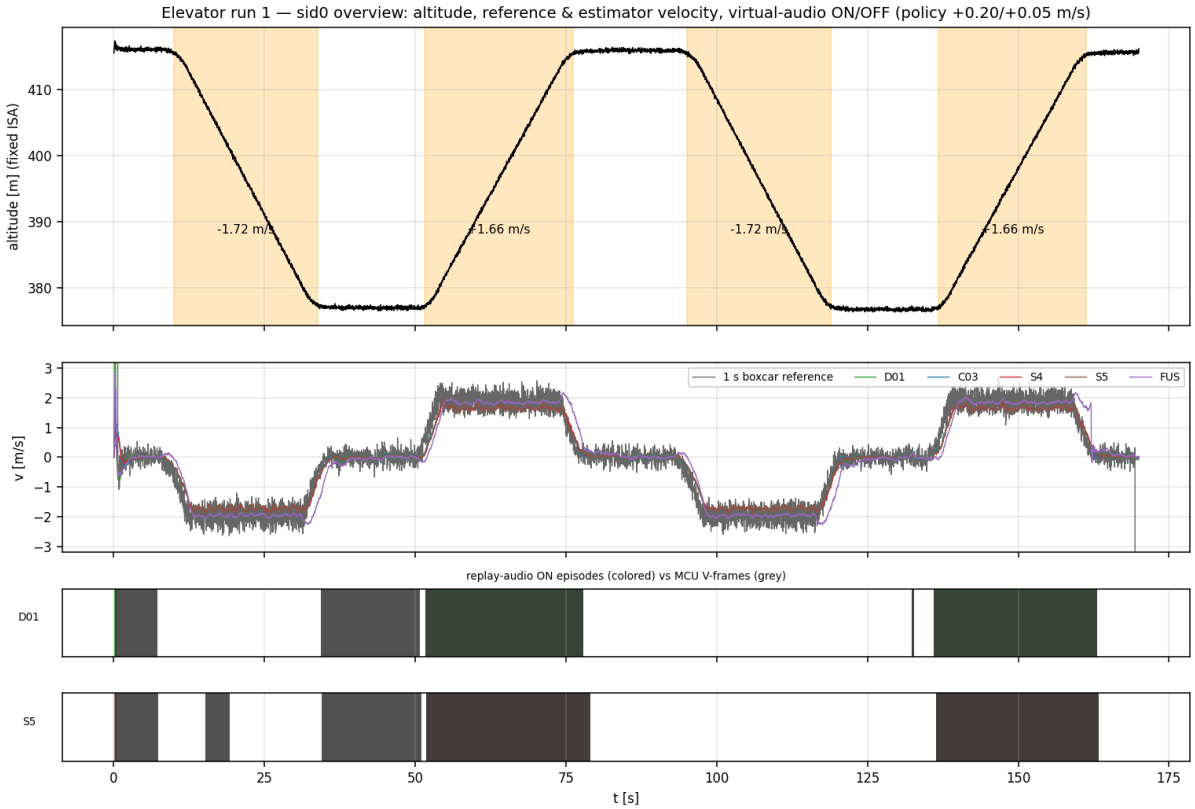


Figure 2. *Run 1 overview: altitude with detected legs, bank velocity traces against the 1-s boxcar reference, and replay-audio ON episodes vs MCU V-frames.*

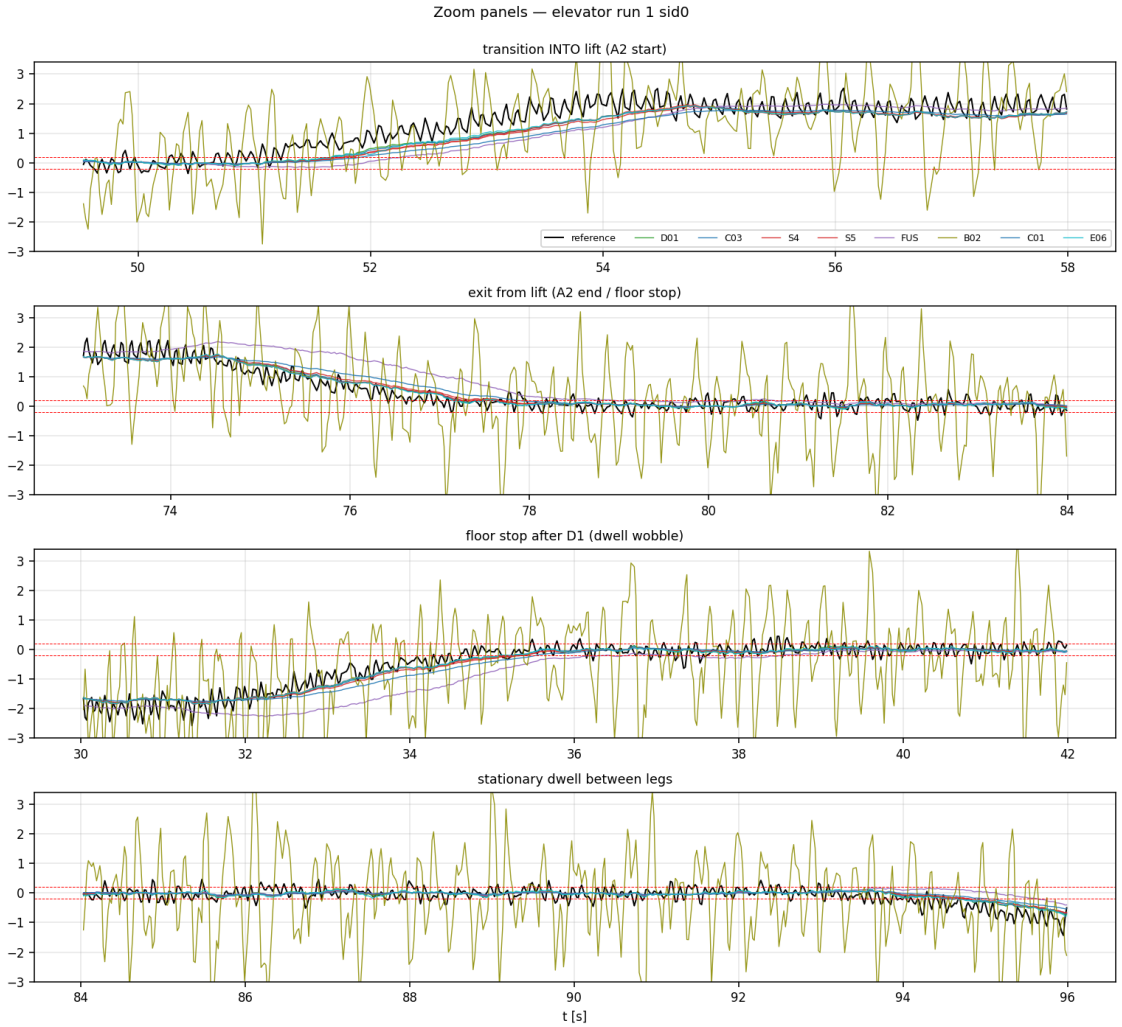


Figure 3. *Zoom panels: transition into lift, exit/floor stop, and dwell behaviour for representative candidates.*

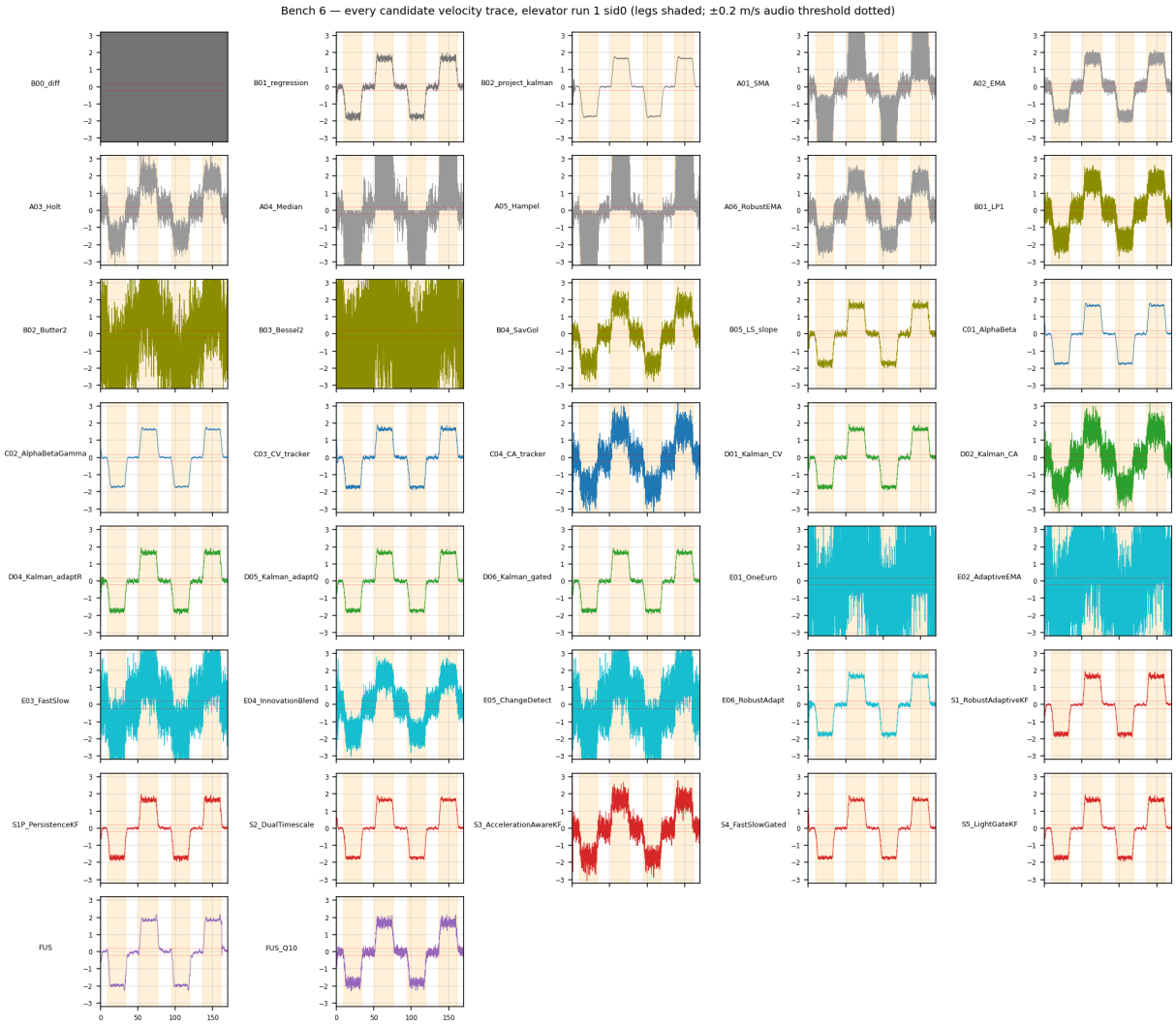


Figure 4. *All 37 candidate velocity traces on run 1 (legs shaded; ±0.2 m/s audio thresholds dotted).*

3.3 Handtest, virtual audio, and CPU

Handtest (corrected reference, DEF-014): the zero-padded reference smoother had fabricated two phantom events (startup 0–3.8 s; toss 148.2–151.9 s, fake ≈4 m/s peak); the IMU refutes them (|a|–g ≤ 0.17 m/s² at 144–152 s — no toss). Event onset/toss metrics are withdrawn. The discriminative handtest metric is slow-block bias: B02 –0.014, C03 –0.030, E06 –0.032, D01 –0.037 m/s (tight); S4 –0.116, S5 –0.121, FUS –0.182; S2 +0.32 and S1 +0.63 m/s — both report climb through a slow descent (S1 audio ON 0.60–0.72 of the block — a slow-descent false-lift). Static-silence false latches (relative): S2 18/h, S1 166/h, bank ≈310–350/h, FUS 629/h.

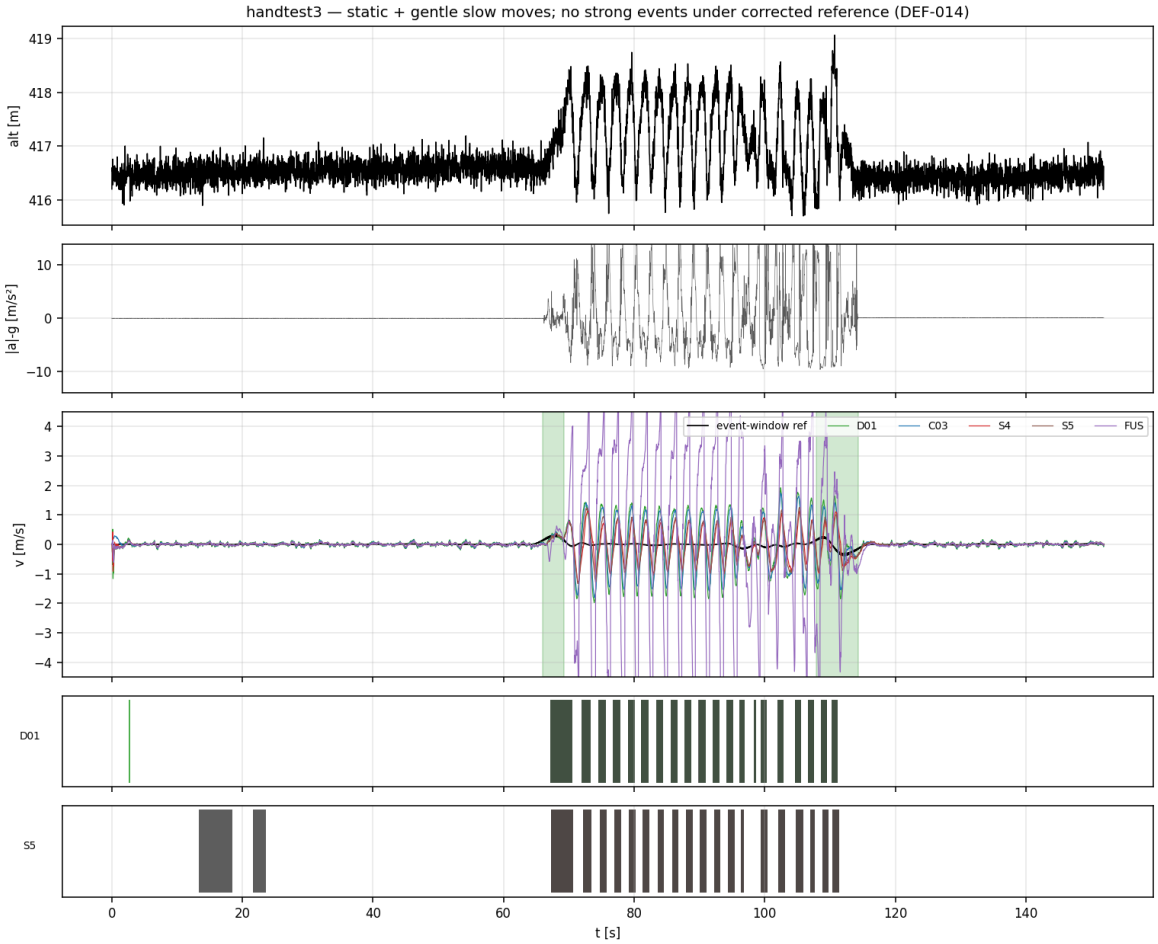


Figure 5. *Handtest3: altitude, IMU acceleration, bank velocity traces with slow-block shading (no events under corrected reference, DEF-014), and virtual-audio overlays.*

Virtual audio, separately: dwell false-ON between legs 2.0–2.4 s/run across the viable family (FUS 4.95) — the post-p0fix residual of the DEF-001/007 mechanism (Phase-era ≈16.5 s/run on the defective reference). Replay-vs-MCU transition agreement (frozen era, ±30 ms window): D01 84 %, S4 43–75 %, S5 33–62 %, dominated by timing near the asymmetric tone clamp rather than state disagreement.

Table 4. *MCU CPU from B-line telemetry during the clean captures (real pipeline, no synthetic timing). Pre-p0fix lines carry the C03 refit storm (~1329/2671 µs, DEF-002 historical evidence). Efficiency is not a differentiator.*

Estimator (on-target, post-p0fix)	sid0 avg/max µs	sid1 avg/max µs	tick budget
D01_Kalman_CV	36 / 44	27 / 37	0.7 %
FUS (baro side; DEF-009)	33 / 37	—	0.7 %
S5_LightGateKF	77 / 80	59 / 88	1.6 %
S4_FastSlowGated	97 / 101	69 / 82	2.0 %
C03_CV_tracker	105 / 132	78 / 102	2.6 %

4. Sampling-rate control

Decimating the clean streams to ≈25/20 Hz and 10 Hz: seconds-parameterized filters (D01, C03, S4, S5, FUS) are near-invariant (leg RMS 0.09→0.13–0.14 at 10 Hz, rise unchanged, exit inside the dwell-wobble floor). The fixed-coefficient B02_Butter2 does not transfer (leg RMS 1.49 m/s at 50 Hz vs 0.32 at 10 Hz) — its Phase 2–5 coefficients are wrong for the ESP-IDF rate; it is excluded from cross-chain comparisons. The ESP-IDF vs Arduino rate differences are therefore a non-issue for the Kalman/tracker/stacked families.

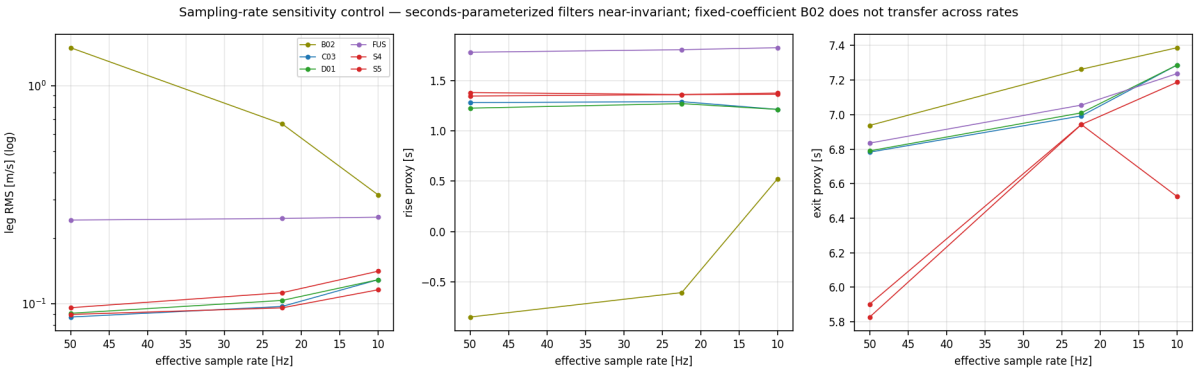


Figure 6. *Rate sensitivity: seconds-parameterized filters near-invariant; fixed-coefficient B02 does not transfer across rates.*

5. Leakage-controlled tuning (NOT a freeze)

Table 5. *Selection on train only (run1 + handtest3); every configuration reported on validation (run2 + statics). The train picks transfer — no leakage signature.*

Family	Train pick	Validation effect vs frozen
D01_Kalman_CV	q=0.3	rest RMS 0.057→0.028, exit 6.8→3.8 s, rise 1.19→1.70 s
S5_LightGateKF	q=0.1	rest RMS 0.037→0.025, exit 5.7→3.7 s, rise 1.33→1.65 s
S4_FastSlowGated	tau_fast=0.5	mixed (exit better, rise/rms worse) — not recommended
C03_CV_tracker	frozen best	all alternatives ≥ +18 % composite

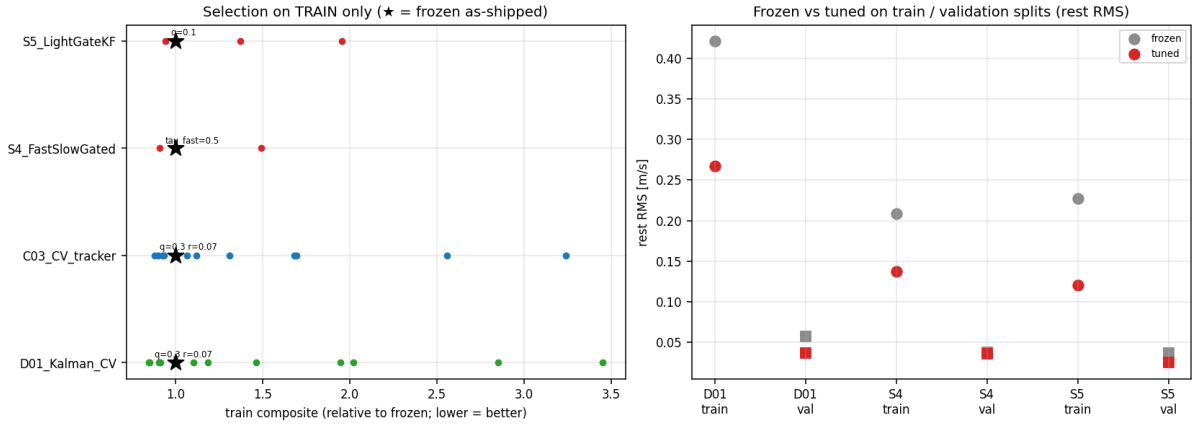


Figure 7. *Tuning pass: train-side selection and train/validation rest-RMS for the frozen vs tuned configurations.*

6. Rankings vs Phases 2–5 and failure modes

Composite (7 metrics, percentile-normalized; slow-block |bias| replaces the phantom onset metric after DEF-014): **C03, C01, B02_project_kalman, E06, S2** top-5; S4 7th, S1 8th, D01 9th, S5 12th, FUS 21st. Per-metric leaders: rest noise B02_project_kalman / C02 / C01; response E06; exit C02/C01 (floored proxy); slow-block tracking B02/E06/C03/D01 (|bias| ≤ 0.04 m/s). **Interpretation flag: S2 is 5th despite its known freeze-trap defect** — the composite rewards its low static chatter; ranking position is not endorsement.

What changed and why: (1) D01's rest floor 0.101→0.057 m/s — the old number was the defective reference, not the filter. (2) S5's Phase-4 stand-out status does not survive corrected streams: the S1/S4/S5 cluster is tight (0.035–0.037) and S5's slow-block bias is mid-pack. (3) S1's defect is now quantified, not shifted: +0.63 m/s slow-descent false-lift — masked in the first analysis by the DEF-014 phantom onset metric. (4) C03 rises to the composite top — its Phase-2/3 weaknesses were firmware-era artefacts (DEF-002 CPU, DEF-001 false-lift), both fixed. (5) FUS shows no measurable benefit on vibration-free data; the fusion value proposition stays open for real-flight data. (6) B02's collapse is a rate artefact.

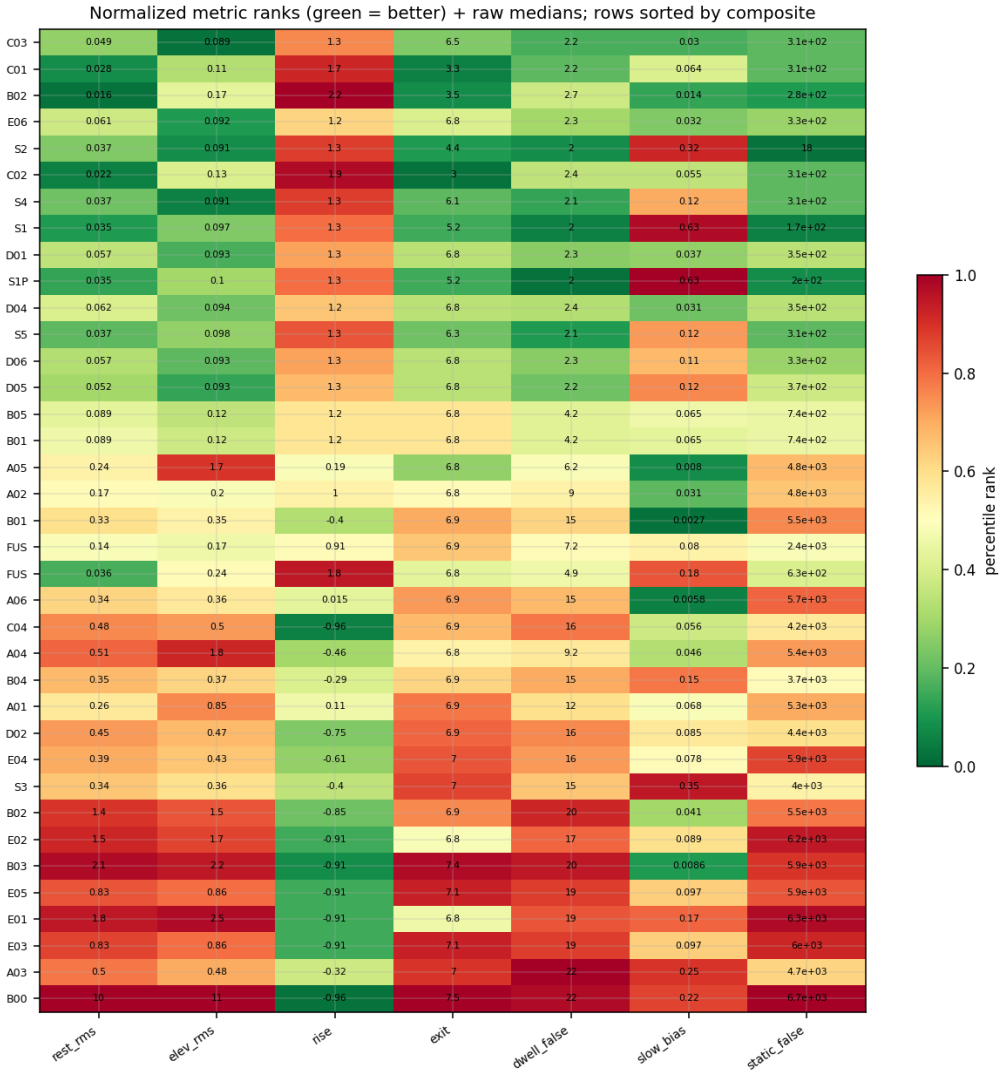


Figure 8. *Normalized metric ranks (rows sorted by composite) with raw medians.*

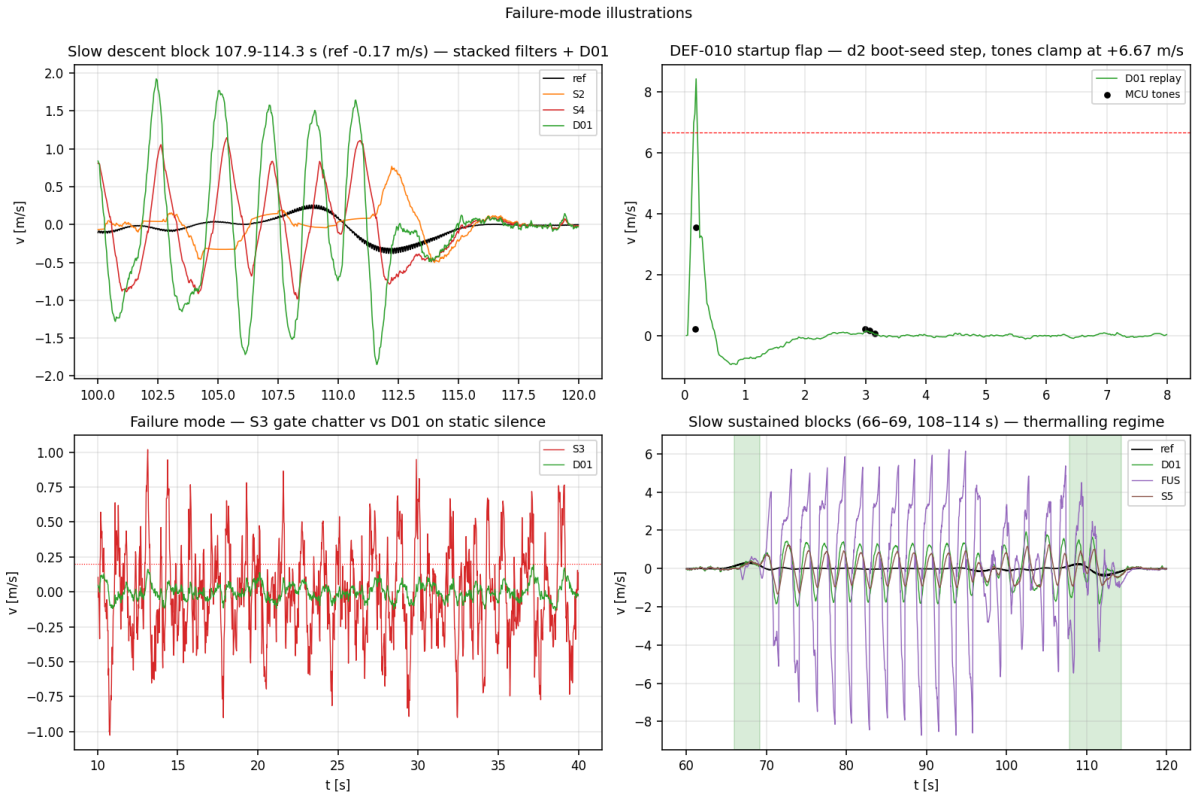


Figure 9. *Failure modes: S1 slow-descent false-lift (surfaced by the DEF-014 correction), S2 freeze-trap, startup flap (DEF-010), S3 gate chatter, slow-block behaviour.*

7. Limitations and next phase

No subsecond ground truth on the elevator: rise/exit are labelled proxies [D]; exit is floored at ≈6.8 s by dwell wobble. Handtest static window contains operator micro-movements; false-latch rates are relative rankings. Post-p0fix dynamic captures do not exist yet — S4/S5/FUS MCU audio equivalence on transitions is vacuous, not failed. Handtest3 has no strong events under the corrected reference (DEF-014); event-relative response latency is not measurable on the current captures. Fusion evaluation is baro-reference-bound. The two clones differ; all medians are reported over both sids.

Next phase (BMP581 + LSM6DSV16X): register the new captures in datasets.py and re-run the pipeline; the registry is the only sensor-specific file. The reference firmware for the current sensor set is archived at esp-idf/firmware_archive/vario_idf_MS5611x2_MPU6050_p0fix_r8e79e68_20261023/ (binaries + SHA-256 + README) for A/B flashing. **Hard gate: DEF-011 (10× pressure scale) must be fixed before any further physical validation, absolute-altitude comparison, or cross-sensor benchmark.** The top-5 MCU efficiency benchmark is deferred until the new sensors arrive (on-target CPU data shows processing is not the present bottleneck).